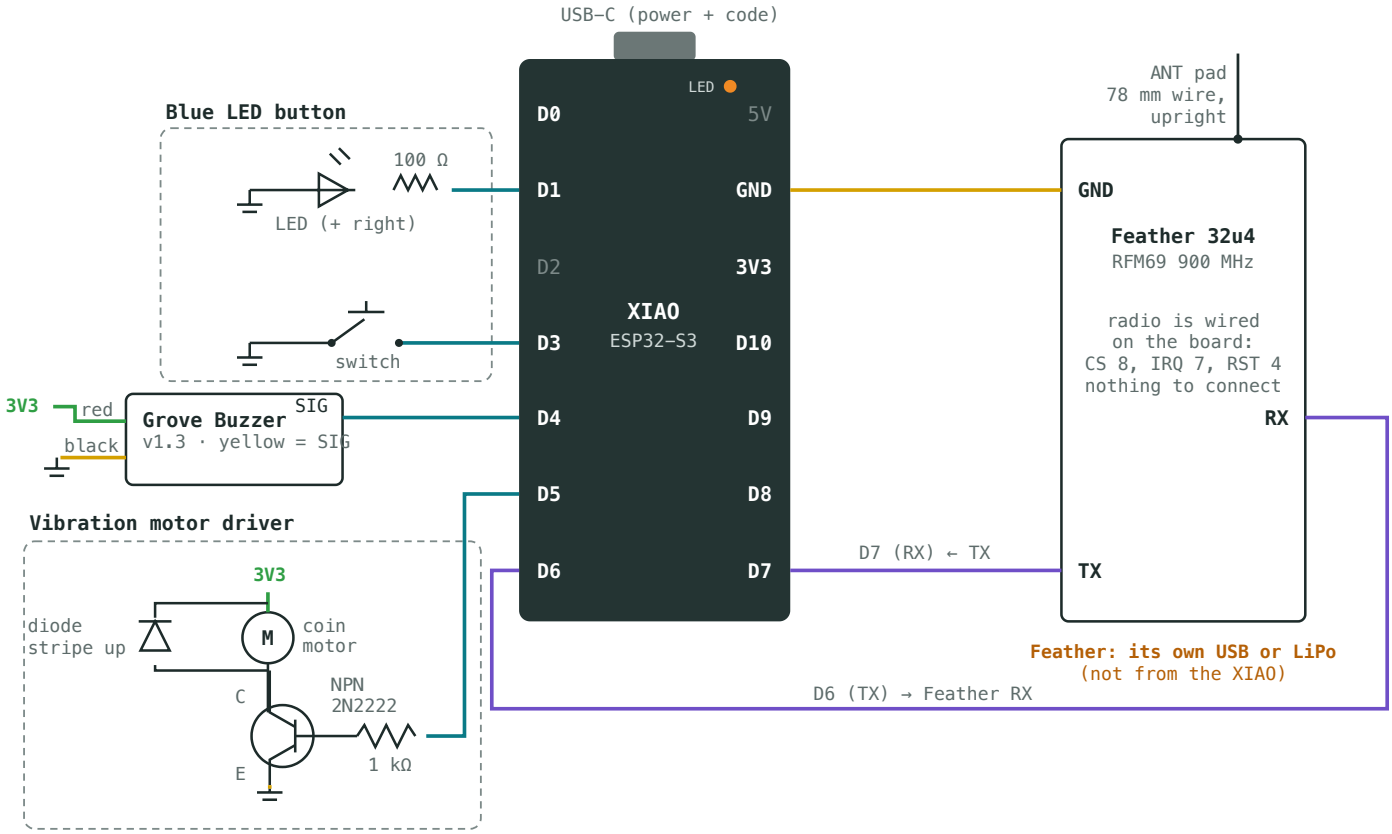
TEAM-11 · IFF BAND · FIRMWARE ENV BAND_XIAO_FEATHER + FEATHER_RADIO

IFF Wristband Wiring

Seeed XIAO ESP32-S3 with a Feather 32u4 RFM69 900 MHz radio (three wires: serial TX, RX and ground), the blue LED button, a Grove buzzer and a coin vibration motor. Pin names are the D labels printed on the XIAO. Unplug USB while you rewire.



3V3 power (green lead) — ground (yellow lead)
serial link to the Feather — control signals

3V3 / ⏚ symbols = the breadboard's 3V3 and GND rails. Team convention: green lead = + (3V3), yellow lead = – (GND).

Wire by wire

FROM	TO	NOTE
Rails		
XIAO 3V3	breadboard + rail (green lead)	everything runs on 3.3 V
XIAO GND	breadboard – rail (yellow lead)	one shared ground for all parts
Feather 32u4 RFM69 radio (serial link, three wires)		
GND	GND rail	common ground: the serial link needs it
RX	**D6**	crossed: the XIAO's TX goes to the Feather's RX
TX	**D7**	and the Feather's TX goes to the XIAO's RX
power	its own USB cable or LiPo	do not power it from the XIAO's pins
ANT	78 mm solid wire	on the Ant. pad; attach before powering on
CS 8, RST 4, IRQ 7	already wired	on the Feather; nothing to connect
Blue LED button (4 legs)		
switch leg 1	**D3**	pull-up is inside the chip

Never wire the motor straight to D5

A coin motor pulls 60 to 100 mA when it starts. A XIAO pin can safely give about 20 mA. The transistor does the switching, and the diode soaks up the spike when the motor stops. The firmware keeps the motor off until you change `PIN_MOTOR` to `D5` in `firmware/include/config.h`.

RFM69 = FSK, not LoRa

The RFM69 has no LoRa mode. This build speaks FSK with channel hopping, so only an FSK node can hear it. Flash the Feather with `pio run -e feather_radio -t upload` (its own USB), the XIAO with `pio run -e band_xiao_feather -t upload`, and the laptop dongle with `pio run -e dongle_fsk -t upload` on an ESP32 + RFM95W. The LoRa dongle and the dog's Feather cannot hear it. Not yet tested on air.

Other radios: a loose RFM69HCW or RFM9x breakout wires straight to the XIAO instead (SCK D7, MISO D8, MOSI D9, CS/NSS D10, RST D0, DIO0 D6; see firmware/README.md). Builds: `band_xiao` (RFM69) and `band_xiao_lora` (RFM9x).

Wire colors

Team leads: **green = + (3V3)**, **yellow = – (GND)**. Modules bring their own colors: the Grove buzzer is red VCC, black GND and **yellow SIG**, and the motor + is red. Go by the pin name, not the color, and check 3V3 and GND before you plug in USB.

FROM	TO	NOTE
switch leg 2	GND rail	pressed = D3 reads LOW
LED +	100 Ω → **D1**	longer leg, or marked +
LED −	GND rail	
Grove Buzzer v1.3 (the module's own wire colors)		
yellow (SIG)	**D4**	tone drive. Yellow here is SIG, not ground
red (VCC)	3V3 rail	
black (GND)	GND rail	white wire unused
Vibration motor driver		
D5	1 kΩ → transistor base	middle leg of a 2N2222
emitter	GND rail	
collector	motor − (blue/black)	
motor + (red)	3V3 rail	
diode	across the motor	stripe to the 3V3 side

Parts you need besides the modules

- 1 × 100 Ω resistor (button LED)
- 1 × 1 kΩ resistor (transistor base)
- 1 × NPN transistor: 2N2222, BC337 or S8050
- 1 × diode: 1N4148 or 1N4001
- 78 mm of solid-core wire for the antenna
- Jumper wires and a breadboard

If your button has 3 wires

Some LED buttons come as a small module with VCC, GND and SIG. Then wire SIG to **D3**, VCC to 3V3 and GND to GND, and skip the 100 Ω resistor. Tell Claude, so it sets `BUTTON_MODULE 1`.

Check it after wiring

1. Plug in USB. The XIAO's LED flashes and the buzzer chirps.
2. The boot log says `#HELLO band radio=ok`. `radio=missing` means the Feather is not answering: check it has power, TX and RX are crossed, and the grounds are joined. Send `#R` to ask it again.
3. Press the button: the log shows `#BTN down` once, and nothing when you leave it alone.
4. Send `#C`: the code 101 plays on the LEDs and buzzer (and motor, once enabled).

Later: battery

The XIAO has BAT+ and BAT− pads underneath for a 3.7 V single-cell LiPo. It charges from USB-C. Mind the polarity: a reversed battery kills the board.